

Section (Check one)

MWF 10am _____

MWF 2pm _____

1. _____/25

2. _____/25

3. _____/25

4. _____/25

TOTAL _____/100

USEFUL INFORMATION

$$\sin(x) = \cos(x - 90^\circ)$$

$$0 < \theta < 180^\circ \text{ (lag)}$$

$$-180^\circ < \theta < 0 \text{ (lead)}$$

$$\bar{V} = \bar{Z}\bar{I} \quad \bar{S} = \bar{V}\bar{I}^* = P + jQ$$

$$I_L = \sqrt{3}I_\phi \text{ (delta)}$$

$$V_L = \sqrt{3}V_\phi \text{ (wye)}$$

$$\bar{Z}_Y = \bar{Z}_\Delta/3$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

$$\bar{S}_{3\phi} = \sqrt{3}V_L I_L \angle \theta$$

ABC phase sequence has A at 0°, B at -120°, and C at +120°

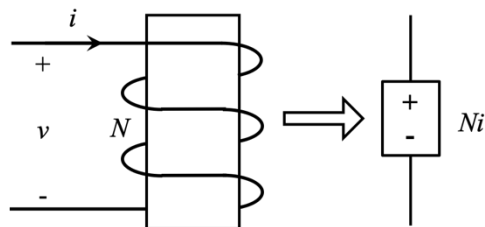
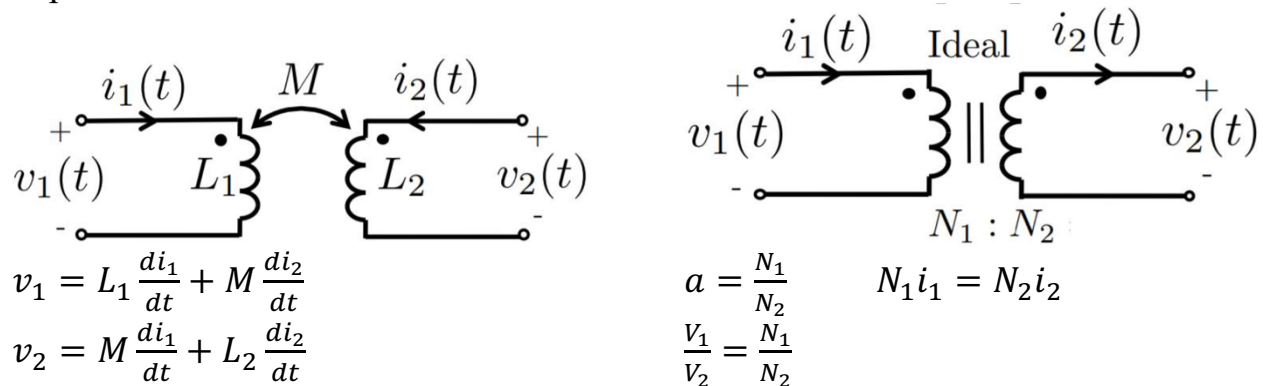
$$\int \underline{H} \cdot \underline{dl} = \int \underline{J_f} \cdot \underline{\hat{n}} dA \quad \int \underline{E} \cdot \underline{dl} = -\frac{d}{dt} \left(\int \underline{B} \cdot \underline{\hat{n}} dA \right) \quad \mathcal{R} = \frac{l}{\mu A} \quad Ni = \mathcal{R}\phi$$

$$\phi = BA$$

$$\lambda = N\phi = Li \text{ (if linear)}$$

$$v = \frac{d\lambda}{dt} \quad k = \frac{M}{\sqrt{L_1 L_2}}$$

$$1 \text{ hp} = 746 \text{ W}$$



Problem 1 (25 Points)

A single phase source is serving three loads connected in parallel, through a feeder with impedance $\bar{Z}_{\text{feeder}} = 1 + j2 \, \Omega$.

- Load 1 is an impedance of $30 + j40 \, \Omega$.
- Load 2 draws 200 W at a lagging PF of 0.8.
- Load 3 draws 1 A at a leading PF of 0.6.

The voltage across the loads is $\bar{V}_{\text{load}} = 100 \angle 0^\circ \text{ V}$.

a) Compute the current phasors $\bar{I}_1$, $\bar{I}_2$, and $\bar{I}_3$ for each of the three loads (20 points total).

c) What is the source voltage $\bar{V}_{\text{source}}$? (3 points)

d) What is the power factor at the source? State whether it is leading or lagging. (1 point)

e) What is the power factor at the load? State whether it is leading or lagging. (1 point)

Problem 2 (25 Points)

A Wye-connected three-phase generator delivers 1200 kVA at 0.6 PF lagging and 4160 volt (line-to-line) to the following loads in parallel.

- Load 1 is connected in Wye, and draws a total of 300 kW at unity power factor.
- Load 2 is an impedance load connected in Delta, with unknown per-phase impedance $\bar{Z}$.

a) Draw the three-phase circuit diagram. Label all relevant instances of the following:

- Phase-to-neutral voltage phasor (e.g. $10\angle 50^\circ$ V).
- Per-phase impedance for load 2 as symbolic expressions (e.g. $\bar{Z}/\sqrt{3}$ Ω).

Include polarity markers (+/-) in all of your voltage labels. (11 points)

b) Draw the single-phase equivalent circuit for phase a. Label one instance of the following:

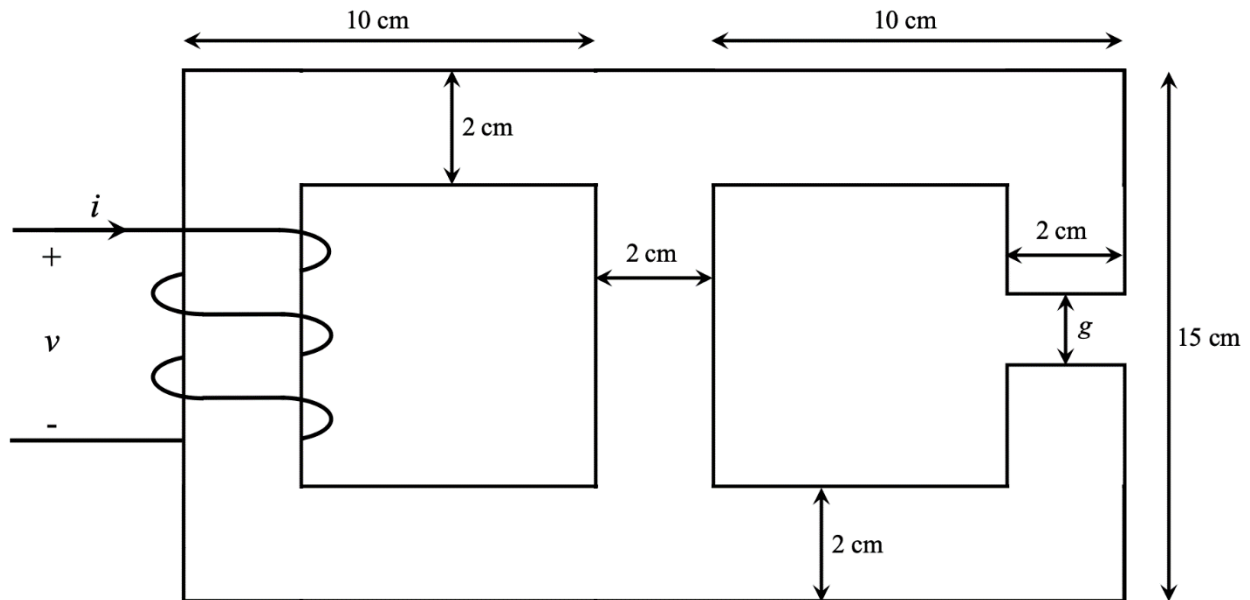
- Voltage phasor (e.g. $10\angle 50^\circ$ V).
- Power for the generator in terms of kVA and PF (e.g. 10 kVA at 0.1 PF leading).
- Power for load 1 in terms of kVA and PF (e.g. 10 kVA at 0.1 PF leading).
- Impedance for load 2 as a symbolic expression (e.g. $\bar{Z}/\sqrt{3}$ Ω).

All quantities should be stated in per-phase values. (6 points)

d) How much three-phase kVARs do we need to bring the combined power factor to unity? How would you connect the capacitors (in Wye or Delta) and why? (3 points)

e) Compute the numerical value for the unknown impedance $\bar{Z}$. (5 points)

Problem 3 (25 Points)



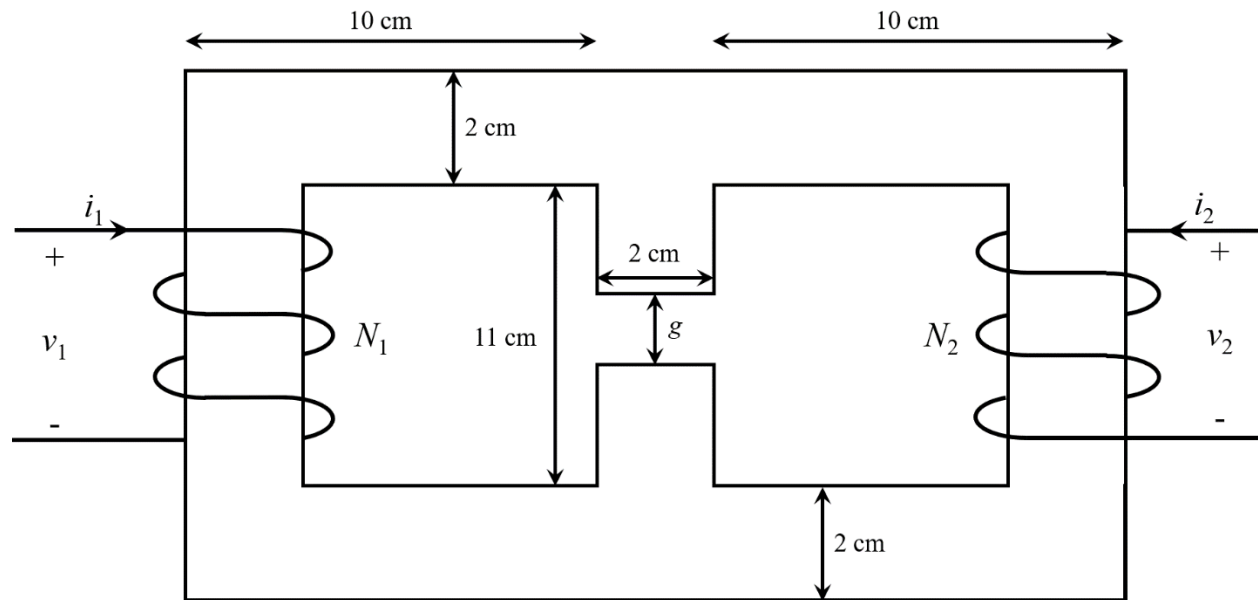
A coil is wrapped 300 times around an iron core with the given dimensions and a depth of 2 cm into the page. The iron has a permeability $\mu=2000\mu_0$ and contains an air gap of $g=2$ mm. Fringing effects can be neglected.

- a) Draw the magnetic path through the iron core on the figure above and draw the corresponding magnetic equivalent circuit. (15 points).

b) What is the total flux ϕ through the circuit? (7 points)

c) What is the inductance of the coil? (3 points)

Problem 4 (25 Points)



Two coils are wrapped around an iron core with the given dimensions and depth of 3 cm into the page. Coil 1 contains 400 turns and Coil 2 contains 200 turns. The iron has a permeability $\mu=1500\mu_0$ and contains an air gap of $g=4$ mm. Fringing effects can be neglected.

- Draw on the figure above where the dot marks should go for each coil. (2 points)
- What is the self-inductance for each coil (L_1 and L_2) and the mutual inductance between the two coils (M)? (18 points)

c) What is the open circuit voltage measured across Coil 2 if a current of $i_1 = \sqrt{2}(10)\cos(377t)$ is applied across Coil 1? Write your answer as a cosine function. (5 points)

(Blank page for extra work)